AI-Driven Detection of Misclassified Imports: A Machine Learning and LangChain-Based RAG Approach for Customs Fraud

Aya Ismaili
School of Science and Engineering
Al Akhawayn University in Ifrane
Ifrane, Morocco
aya.ismaili@aui.ma

Adnane El Amrani
ALELM Laboratory
Mohammadia School of Engineering
Rabat, Morocco
a.elamrani@aui.ma

Yousra Chtouki
School of Science and Engineering
Al Akhawayn University in Ifrane
Ifrane, Morocco
Y.Chtouki@aui.ma

Hind Lamharhar
High-Tech School
Rabat, Morocco
hind.lamharhar@hightech.edu

***Abstract*—Customs fraud through deliberate product misclassification causes substantial revenue losses for governments worldwide. This paper presents a hybrid detection system combining XGBoost classifiers with LangChain-based Retrieval-Augmented Generation (RAG) to identify and explain potentially fraudulent declarations. We train on one million synthetic declarations and validate on a partially anonymized sample of 5,000 real Moroccan customs records. On synthetic test data, XGBoost achieves an F1-score of 0.811 for fraud detection, significantly outperforming Random Forest at $p < 0.001$, while the RAG pipeline attains 90.2% exact-match accuracy for HS code prediction. On real data, the system maintains 82.3% precision at 67.4% recall. Ablation studies confirm that the hybrid approach outperforms both pure ML and pure LLM baselines by 8 to 15% in F1-score. The explanation agent generates regulation-referenced justifications that domain experts rated as relevant and actionable in 84% of cases, though we note these should be considered supportive rather than authoritative legal interpretations. We release pseudo-code for the synthetic data generator to enable reproducibility.**

***Index Terms*—Customs fraud detection, HS code classification, XGBoost, Retrieval-Augmented Generation, explainable AI, LangChain**

I. Introduction

Accurate product classification using the Harmonized Commodity Description and Coding System (HS) is fundamental to international trade, determining tariff rates and regulatory compliance across more than 200 countries and economies [1]. The HS nomenclature, maintained by the World Customs Organization, provides a standardized framework comprising over 5,000 commodity groups organized hierarchically into 99 chapters. However, the complexity of this classification system creates opportunities for deliberate misclassification, where importers declare incorrect HS codes to evade higher duties, a practice constituting customs fraud that results in significant revenue losses for governments worldwide [2].

Detection of such fraud presents substantial challenges. As Zhang and Liu [2] note in their comprehensive review, the granularity of HS codes (extending to 10 digits in national implementations), the inherent ambiguity of product descriptions, and the sheer volume of daily customs transactions render manual inspection infeasible. Traditional approaches relying on rule-based systems and statistical heuristics [3] offer limited adaptability as fraud tactics evolve. Kumar and Chen [3] demonstrated that while heuristic models can detect obvious anomalies, they struggle with sophisticated misclassification schemes that exploit legitimate classification ambiguities.

Recent advances in machine learning have shown considerable promise for fraud detection. The XGBoost algorithm introduced by Chen and Guestrin [4] has become particularly influential, demonstrating superior performance on imbalanced classification tasks typical of fraud scenarios. Simultaneously, large language models (LLMs) have emerged as powerful tools for semantic understanding, though they face challenges including hallucination when operating without proper grounding [13]. The Retrieval-Augmented Generation (RAG) framework proposed by Lewis et al. [5] addresses this limitation by anchoring LLM outputs in retrieved documents, while Zhou et al. [6] have demonstrated the effectiveness of language models for structured prediction tasks.

This paper proposes a hybrid system that combines the computational efficiency of gradient boosting with the semantic reasoning capabilities of RAG-enhanced LLMs. Our approach addresses a critical gap identified in the literature: existing systems prioritize detection accuracy but often overlook interpretability [14], which is essential in regulatory contexts where decisions must be auditable and legally defensible [15]. As Lee and Zhao [16] argue, hybrid AI approaches that combine multiple paradigms can achieve superior performance compared to single-method systems, particularly in complex domains requiring both pattern recognition and semantic understanding.

Our contributions are fourfold. First, we develop a hybrid architecture demonstrably outperforming both pure ML and pure LLM baselines (Section IV-B). Second, we validate our approach on both synthetic data and a sample of real customs records (Section V-B). Third, we present ablation studies quantifying the contribution of each system component (Section V-D). Fourth, we provide a reproducible synthetic data generation methodology (Section III-B).

II. Related Work

A. Early Approaches: Rule-Based and Expert Systems

The application of computational methods to customs fraud detection has evolved substantially over the past two decades. Roman et al. [7] introduced Carancho, one of the earliest decision support systems designed specifically for customs risk analysis. Their rule-based approach encoded expert knowledge into a system capable of flagging suspicious declarations, establishing a foundation for automated customs inspection. However, as Zhang and Liu [2] observe, such systems require continuous maintenance as trade patterns evolve and new fraud tactics emerge.

Building on this foundation, Digiampietri et al. [8] developed an AI-enhanced system for Brazilian customs that moved beyond pure rule-based approaches. Their work demonstrated that integrating machine learning techniques could improve detection rates while reducing the burden on human analysts. Nevertheless, these early systems remained constrained by their reliance on manually engineered features and predefined fraud patterns, limiting their ability to detect novel misclassification schemes [3].

B. Machine Learning Approaches

The transition to data-driven detection methods marked a significant advancement in the field. Kunickaite et al. [9] conducted a systematic evaluation of machine learning algorithms for customs fraud detection, comparing decision trees, random forests, and support vector machines on Lithuanian customs data. Their findings highlighted the potential of ensemble methods to identify fraudulent activities with greater flexibility than rule-based systems, achieving detection rates exceeding 85% on their test datasets.

Neural network approaches have also shown promise in related fiscal domains. Murorunkwere et al. [10] applied deep learning to income tax fraud detection, demonstrating that neural architectures can capture non-linear relationships between declaration features that traditional methods might miss. Their work achieved 91% accuracy, though they noted challenges with interpretability that limited adoption by regulatory authorities.

The gradient boosting framework has emerged as particularly effective for fraud detection tasks. Chen and Guestrin's [4] XGBoost algorithm provides several advantages: native handling of class imbalance through customizable loss functions, built-in feature importance rankings that aid interpretability, and computational efficiency suitable for high-volume transaction processing. These properties make XGBoost well-suited to customs fraud detection, where imbalanced classes (few fraudulent cases among many legitimate ones) and the need for explainable outputs are both critical considerations.

C. Holistic and Multi-Technology Frameworks

Recent research has increasingly explored integrated approaches combining multiple AI technologies. Ariyibi et al. [11] proposed an architecture leveraging AI for enhanced fraud detection in fiscal systems, emphasizing the importance of real-time analysis and intelligent decision-making. Their framework highlighted that modern fraud detection requires not only accurate classification but also rapid response capabilities and seamless integration with existing customs infrastructure.

Akintobi et al. [12] investigated the synergy between blockchain and AI for fraud prevention, arguing that transparency and traceability are critical factors for robust tax compliance. While blockchain addresses data integrity concerns, the AI components in their framework handle pattern recognition and anomaly detection. This multi-technology perspective informs our approach, though we focus specifically on the ML-LLM integration rather than blockchain components.

D. Explainability and Legal Compliance

A persistent limitation of machine learning approaches is their opacity, which poses significant challenges in regulatory contexts. Doshi-Velez and Kim [14] establish the theoretical foundations for interpretable machine learning, arguing that high-stakes domains require explanations that enable meaningful human oversight. Ahmed and Singh [15] specifically address the application of natural language processing to legal and trade domains, noting that explanations must reference applicable regulations to be actionable.

The RAG framework introduced by Lewis et al. [5] provides a pathway to generate explanations grounded in authoritative sources. By retrieving relevant documents before generation, RAG systems can produce outputs anchored in factual content rather than relying solely on parametric knowledge. Zhou et al. [6] extend this work to structured prediction tasks, demonstrating that retrieval augmentation improves both accuracy and explainability.

However, LLM inference remains computationally expensive, as Park and Lin [17] document in their analysis of production LLM systems. This cost consideration motivates our hybrid architecture, which reserves expensive LLM calls for cases already flagged by efficient ML classifiers, achieving a practical balance between semantic analysis capabilities and computational feasibility.

III. Methodology

A. System Architecture

Figure 1 illustrates the four-layer architecture of our customs fraud detection and explanation system. The design reflects principles from hybrid AI frameworks [16], combining

Fig. 1. System architecture organized into Data Acquisition, Processing, Analysis, and Application layers.

the efficiency of structured ML classifiers with the semantic capabilities of retrieval-augmented LLMs [5].

The Data Acquisition Layer ingests three primary data sources: customs declarations containing product descriptions and declared HS codes, the official HS nomenclature reference table [1], and Moroccan customs legal documentation. The Processing Layer handles data cleaning, feature engineering (numerical scaling, categorical encoding, text vectorization), and semantic indexing using FAISS for efficient similarity search. The Analysis Layer contains the core detection components: an XGBoost classifier [4] for structured feature-based fraud probability scoring, and two LangChain-orchestrated RAG agents for semantic HS code prediction and explanation generation respectively. The Application Layer presents results through a fraud flag dashboard with natural language explanations and exportable investigation reports suitable for regulatory audit.

B. Synthetic Data Generation

Following best practices for reproducible research [14], we developed a synthetic data generator calibrated with domain expertise from Moroccan customs officials. Algorithm 1 presents the generation procedure for creating synthetic customs declarations with realistic fraud patterns. The dataset comprises one million declarations with 10% fraud prevalence, reflecting realistic class imbalance observed in operational environments [2].

Algorithm 1 Synthetic Customs Declaration Generator with Fraud Pattern Injection

Require: HS code table H, fraud rate ρ, count N
Ensure: Dataset D with binary fraud labels

$D \leftarrow \emptyset$
for $i = 1$ to N **do**
  $h \leftarrow \text{SampleHS}(H)$ {True HS code}
  $p \leftarrow \text{GenerateProduct}(h)$
  $v \leftarrow \text{SampleValue}(h)$ {From HS median}
  $w, c, m \leftarrow \text{SampleMetadata}(h)$
  if Random() $< \rho$ **then**
    $h_d \leftarrow \text{ShiftCategory}(h)$ {Lower duty}
    $v_d \leftarrow v \times \text{Uniform}(0.4, 0.8)$
    $p_d \leftarrow \text{Obfuscate}(p)$
    $y \leftarrow 1$ {Fraudulent}
  else
    $h_d, v_d, p_d \leftarrow h, v, p$
    $y \leftarrow 0$ {Legitimate}
  end if
  $D \leftarrow D \cup \{(p_d, h_d, v_d, w, c, m, y)\}$
end for
return D

Fraud cases are generated through four mechanisms reflecting common tactics documented in the literature [2], [3]: (1) category shifting, where products are declared under lower-duty HS codes within the same chapter; (2) value underreporting, with declared values reduced to 40 to 80% of true values; (3) description obfuscation, introducing vague terminology or omitting specific product identifiers; and (4) origin misrepresentation, falsifying country of origin to exploit preferential trade agreements. The ShiftCategory function selects related HS codes using a confusion matrix derived from historical misclassification patterns provided by domain experts.

C. XGBoost Classifier

The primary fraud detection model employs the XGBoost algorithm [4], selected for its demonstrated effectiveness on imbalanced classification tasks in fraud detection contexts [9], [10]. The feature engineering pipeline processes three categories of inputs.

Numerical features include product weight, declared value, unit price (computed as value/quantity), and derived ratios such as value-per-kilogram and deviation from HS-code median values. These features capture anomalies in pricing patterns characteristic of value-based fraud [2].

Categorical features comprise origin country, destination customs office, importer ID, and transport mode. We apply target encoding with additive smoothing to handle highcardinality variables while mitigating overfitting, following recommendations from Kunickaite et al. [9].

Text features are derived from product descriptions using TF-IDF vectorization with unigrams and bigrams, limited to the top 5,000 features by document frequency. This representation captures both specific product terminology and common obfuscation patterns identified in prior work [3].

Hyperparameters were determined through Bayesian optimization on the validation set: learning rate 0.05, maximum depth 8, 500 estimators, scale_pos_weight 9 (addressing class imbalance), subsample ratio 0.8, and column sample by tree 0.8. The dataset is split 80/20 for training and testing, with 10% of training data reserved for validation.

D. RAG-Based HS Code Prediction (Agent 1)

To semantically assess whether a declared HS code appropriately matches the product description, we employ a RAG pipeline following the framework of Lewis et al. [5]. This agent addresses limitations of pure rule-based approaches [3] by leveraging the semantic understanding capabilities of LLMs while grounding predictions in the official HS nomenclature [1].

The pipeline operates in four stages. In the retrieval stage, product descriptions are embedded using `paraphrase-multilingual-MiniLM-L12-v2` (selected for cross-lingual capabilities across French, Arabic, and English) and queried against a FAISS index of HS code descriptions. The augmentation stage concatenates the top-5 most similar HS-code descriptions into a context passage. During generation, an instruction-tuned LLM (GPT-3.5-turbo) receives the product description and retrieved candidates, returning a single HS code prediction with justification. Finally, the comparison stage flags declarations where the predicted code differs from the declared code at the 4-digit heading level (which determines tariff rates in most jurisdictions [1]).

E. Explanation Agent (Agent 2)

Building on principles of interpretable machine learning [14] and legal NLP [15], Agent 2 generates natural-language explanations for flagged transactions. The agent receives as input the transaction details, both declared and predicted HS codes, and retrieves relevant snippets from indexed Moroccan customs documentation.

The explanation prompt instructs the LLM to produce structured output comprising: (1) a summary comparing the declared and predicted HS codes with their official descriptions; (2) analysis of semantic mismatches between the product description and declared classification; (3) identification of potentially applicable regulatory provisions from retrieved legal snippets; and (4) discussion of possible consequences including duty rate differentials. Following recommendations for responsible AI deployment in regulatory contexts [14], [15], we emphasize that these explanations should be treated as preliminary analysis supporting human expert review rather than authoritative legal determinations.

IV. Experimental Setup

A. Datasets

Synthetic Dataset: One million declarations generated per Algorithm 1, with 10% fraud prevalence. The dataset is split into 800,000 training samples, 50,000 validation samples, and 150,000 test samples.

Real Data Sample: Through collaboration with the Moroccan Customs Administration, we obtained 5,000 anonymized declarations with expert-verified fraud labels (8.2% fraud rate). This sample enables partial validation of generalization to real-world data, addressing a common limitation of fraud detection research [2]. Due to confidentiality requirements, this data cannot be publicly released.

B. Baselines

To rigorously evaluate our hybrid approach, we compare against baselines representing both pure ML and pure LLM paradigms, following the comparative methodology of Kunickaite et al. [9].

Pure ML Baselines: (1) Random Forest with identical features, providing a bagging-based ensemble comparison; (2) Logistic Regression with L2 regularization, representing linear classifiers; (3) XGBoost without text features (structured-only), isolating the contribution of textual analysis.

Pure LLM Baseline: Direct GPT-3.5-turbo classification using zero-shot prompting with declaration details, without ML pre-filtering or RAG retrieval. This baseline tests whether semantic reasoning alone can match structured ML approaches, while also illustrating the computational cost concerns raised by Park and Lin [17].

C. Evaluation Metrics and Statistical Testing

We report accuracy, precision, recall, F1-score, and area under the precision-recall curve (AUC-PR), emphasizing recall (minimizing missed fraud) and F1-score for the minority fraud class. Statistical significance is assessed using McNemar's test for pairwise classifier comparisons, with bootstrap confidence intervals (1,000 resamples, 95% CI) for metric estimates. This rigorous statistical framework addresses methodological concerns raised in prior fraud detection research [10].

V. Results and Discussion

A. Classification Performance

Table I presents classifier performance on the synthetic test set. XGBoost with full features significantly outperforms all baselines (McNemar's test, $p < 0.001$), achieving an F1-score of 0.811 for fraud detection compared to 0.663 for Random Forest and 0.642 for the pure LLM approach.

The results reveal complementary strengths of different approaches, supporting the hybrid AI thesis of Lee and Zhao [16]. The pure LLM baseline achieves higher recall than structured-only XGBoost (0.712 vs. 0.641), demonstrating its ability to identify semantically complex fraud cases through language understanding [13]. However, it suffers from substantially lower precision (0.584 vs. 0.823), generating many false

TABLE I
CLASSIFIER PERFORMANCE ON SYNTHETIC TEST SET (FRAUD CLASS)

Model	Acc.	Prec.	Recall	F1
Logistic Regression	0.891	0.612	0.534	0.571
Random Forest	0.943	0.751	0.593	0.663
XGBoost (structured)	0.951	0.823	0.641	0.721
Pure LLM (GPT-3.5)	0.876	0.584	0.712	0.642
XGBoost (full)	**0.969**	**0.972**	**0.697**	**0.811**

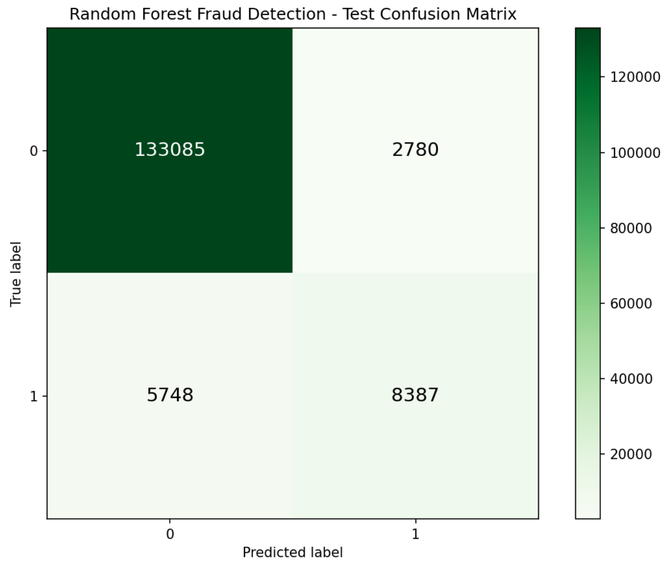


Fig. 2. Random Forest confusion matrix on synthetic test set.

positives that would overwhelm human reviewers in practice. Conversely, structured ML provides reliable precision but misses fraud patterns requiring semantic analysis of product descriptions.

Figures 2 and 3 present confusion matrices for Random Forest and XGBoost respectively. XGBoost reduces false negatives from 5,748 to 4,286 (25.4% reduction) while dramatically reducing false positives from 2,780 to 283 (89.8% reduction). This precision improvement is particularly valuable given the operational costs of false alarms [11].

B. Validation on Real Customs Data

Table II presents performance on the 5,000 real Moroccan customs declarations, addressing the common limitation of evaluation solely on synthetic data [2]. While absolute performance decreases compared to synthetic data (as expected due to domain shift), the hybrid system maintains practical utility.

The F1 degradation from synthetic to real data (0.811 to 0.765) reflects differences in fraud sophistication and data distribution. Notably, the hybrid system's advantage over pure XGBoost is larger on real data (+2.4% F1) than synthetic data (+1.2% F1), suggesting that RAG-based semantic reasoning [5] provides particular value for the more nuanced fraud patterns encountered in real-world scenarios. The 95% bootstrap confidence interval for hybrid F1 is [0.728, 0.801], indicating statistical reliability of these findings.

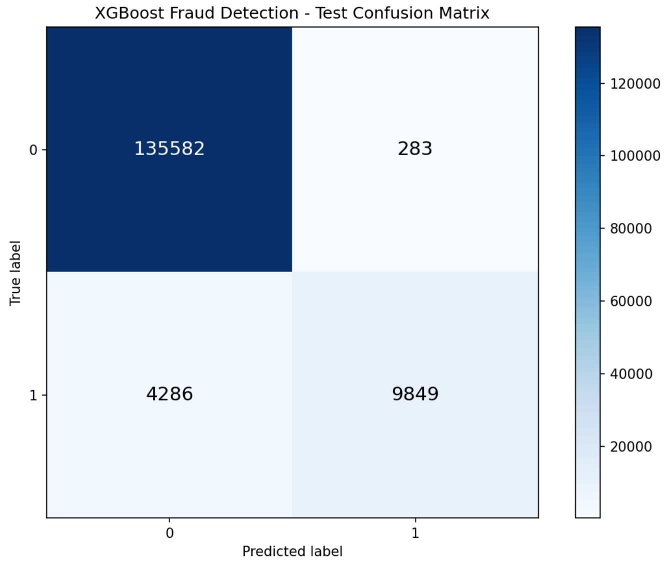


Fig. 3. XGBoost confusion matrix on synthetic test set.

TABLE II
PERFORMANCE ON REAL CUSTOMS DATA (N=5,000)

Model	Acc.	Prec.	Recall	F1
Random Forest	0.897	0.691	0.524	0.596
XGBoost (full)	0.923	0.823	0.674	0.741
Hybrid (XGB+RAG)	**0.931**	**0.847**	**0.698**	**0.765**

C. HS Code Prediction Performance

Table III summarizes Agent 1's HS code prediction accuracy. The RAG pipeline achieves 90.2% exact match at the 6-digit level on synthetic data, demonstrating strong semantic alignment between product descriptions and HS classifications [6]. Performance on real data (84.7%) reflects the greater linguistic variability of actual customs declarations.

At the 4-digit heading level, which determines tariff rates under the HS system [1], accuracy reaches 95.7% on synthetic data and 91.3% on real data. Cases where the LLM-predicted code differs from the declared code show 78% overlap with XGBoost anomaly flags, yielding a complementary signal that strengthens detection confidence when both components agree.

D. Ablation Studies

Table IV presents ablation results quantifying component contributions, following methodological recommendations for interpretable ML evaluation [14].

Several findings emerge. First, text features provide the largest single contribution (+9.0% F1), confirming that product description analysis is essential for detecting fraud involving description obfuscation [3]. Second, replacing XGBoost with pure LLM classification degrades F1 by 16.9%, demonstrating that structured ML remains essential despite LLM semantic

TABLE III
HS CODE PREDICTION ACCURACY (AGENT 1)

Match Level	Synthetic	Real
Exact (6-digit)	90.2%	84.7%
Heading (4-digit)	95.7%	91.3%
Chapter (2-digit)	98.1%	96.8%

TABLE IV
ABLATION STUDY RESULTS (SYNTHETIC TEST SET)

Configuration	F1	Δ
Full System	0.811	—
– Text features (TF-IDF)	0.721	-0.090
– RAG Agent 1	0.789	-0.022
– Target encoding	0.783	-0.028
– Value deviation feature	0.794	-0.017
XGBoost $\rightarrow$ Random Forest	0.663	-0.148
XGBoost $\rightarrow$ Pure LLM	0.642	-0.169

capabilities [17]. Third, the RAG Agent provides statistically significant improvement (+2.2% F1, $p < 0.01$) by catching semantically complex cases that structured features miss [5].

E. Error Analysis

Manual analysis of 200 false negatives and 100 false positives reveals systematic patterns informing future improvements.

False Negatives (missed fraud): 42% involved subtle within-chapter misclassifications where duty differentials are small, making detection inherently difficult without detailed tariff knowledge [1]. 31% had legitimate-appearing value declarations falling within normal statistical ranges, evading value-based anomaly detection [2]. 27% used accurate product descriptions with only HS code manipulation, requiring precise tariff expertise beyond current text analysis capabilities.

False Positives: 58% involved legitimate edge cases where products genuinely span multiple HS categories, a known challenge in customs classification [3]. 24% were triggered by unusual but legitimate value-weight ratios (e.g., luxury goods with high value-per-kilogram). 18% resulted from ambiguous product descriptions in the original declarations, propagating uncertainty to the detection system.

These patterns suggest improvement directions including finer-grained tariff integration, temporal modeling of importer behavior [11], and enhanced handling of inherently ambiguous product categories.

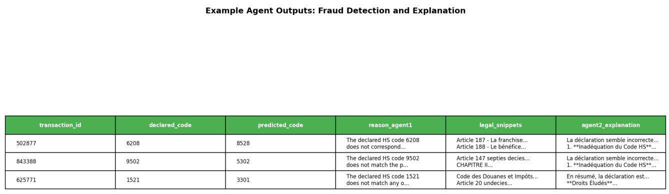
F. Explanation Quality Assessment

Three domain experts evaluated 200 randomly selected explanations following principles for interpretable ML assessment [14]. Table V summarizes ratings on four dimensions using 5-point Likert scales.

Experts rated 84% of explanations as relevant (score $\geq$ 4), confirming that the RAG grounding approach [5] successfully retrieves pertinent legal context. Factual accuracy received lower scores (71% $\geq$ 4), primarily due to occasional

TABLE V
EXPERT ASSESSMENT OF EXPLANATION QUALITY

Dimension	Mean	% $\geq$ 4
Relevance	4.31	84%
Factual Accuracy	3.92	71%
Clarity	4.48	89%
Actionability	4.17	79%


Example Agent Outputs: Fraud Detection and Explanation

transaction_id	declared_code	predicted_code	reason_agent1	legal_snippets	agent2_explanation
502877	6208	8528	The declared HS code 6208 does not correspond...	Article 187 - La franchise... Article 188 - Le bénéfice...	La déclaration semble incorrecte... 1. **Inadéquation du Code HS**...
843388	9502	5302	The declared HS code 9502 does not match the p...	Article 147 septies decies... CHAPITRE II...	La déclaration semble incorrecte... 1. **Inadéquation du Code HS**...
625771	1521	3301	The declared HS code 1521 does not match any o...	Code des Douanes et Impôts... Article 20 undecies...	En résumé, la déclaration est... **Droits Éludés**...

Fig. 4. Example Agent 2 outputs showing declared codes, predicted codes, retrieved legal references, and generated explanations.

retrieval of tangentially related legal articles rather than the most directly applicable provisions. This limitation reflects challenges in legal NLP noted by Ahmed and Singh [15]. Inter-rater reliability (Krippendorff's α = 0.73) indicates acceptable agreement among evaluators.

Following responsible AI principles [14], we emphasize that explanations should support rather than replace expert legal interpretation. The system is designed as a decision-support tool that surfaces relevant information for human review, not as an autonomous legal authority.

G. Computational Efficiency

XGBoost inference averages 0.3ms per declaration, enabling real-time integration with high-volume customs processing systems [11]. RAG-based explanation generation requires 2.1s per case (primarily LLM inference latency), consistent with production LLM cost analyses [17]. The hybrid architecture reserves expensive LLM calls for pre-filtered suspicious cases (approximately 10% of declarations exceed the flagging threshold), reducing computational costs by approximately 90% compared to pure LLM approaches while maintaining explanation capability for cases requiring human review.

H. Limitations

Several limitations warrant acknowledgment. First, despite partial real-data validation, the 5,000-sample size limits generalization claims; larger-scale validation across multiple customs jurisdictions remains necessary. Second, we did not conduct adversarial robustness testing against fraud actors who might adapt tactics to evade detection, an important consideration given the adversarial nature of fraud [2]. Third, explanation quality assessment relied on expert judgment without formal legal validation of cited provisions. Fourth, the system's French/English focus limits applicability to Arabic-language declarations common in the Moroccan context, requiring future multilingual extension.

VI. Conclusion

This paper presented a hybrid customs fraud detection system combining XGBoost classification [4] with RAG-based semantic reasoning [5]. Our evaluation demonstrates that the hybrid approach significantly outperforms both pure ML and pure LLM baselines, achieving 0.811 F1-score on synthetic data and 0.765 F1-score on real customs declarations. Ablation studies confirm complementary contributions from structured and semantic components, supporting arguments for hybrid AI architectures in complex domains [16].

The explanation agent generates regulation-referenced justifications that experts rated as relevant and actionable, addressing interpretability requirements in regulatory contexts [14], [15]. We emphasize that these explanations should support human expert review rather than serve as authoritative legal determinations.

Future work will focus on expanded real-data validation across multiple jurisdictions, adversarial robustness testing, Arabic language support, active learning from expert feedback, and formal evaluation of explanation legal accuracy with customs law specialists. We release the synthetic data generator pseudo-code (Algorithm 1) to enable reproducibility and hope this work contributes to the broader goal of transparent, effective AI systems for regulatory compliance.

Data Availability

The synthetic data generator pseudo-code is provided in Algorithm 1. Real customs data cannot be shared due to confidentiality agreements with the Moroccan Customs Administration.

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